

ELECTROSTATICS

Type of CPP - Objective & Subjective

Level - MODERATE

1. A positive point charge Q is brought ~~Q is brought~~ near a neutral metal cube :
- (A) The cube becomes negatively charged
 (B) The cube becomes positively charged
 (C) The interior becomes positively charged and the surface becomes negatively charged.
 (D) The interior remains neutral and the surface gets nonuniform charge distribution.

Apply gauss law for inside

2. Two conducting plates X and Y , each having large surface area ' A ' as shown in figure (on one side) are placed parallel to each other. The plate X is given a charge Q whereas the other is neutral. The electric field at a point in between the plates is given by :

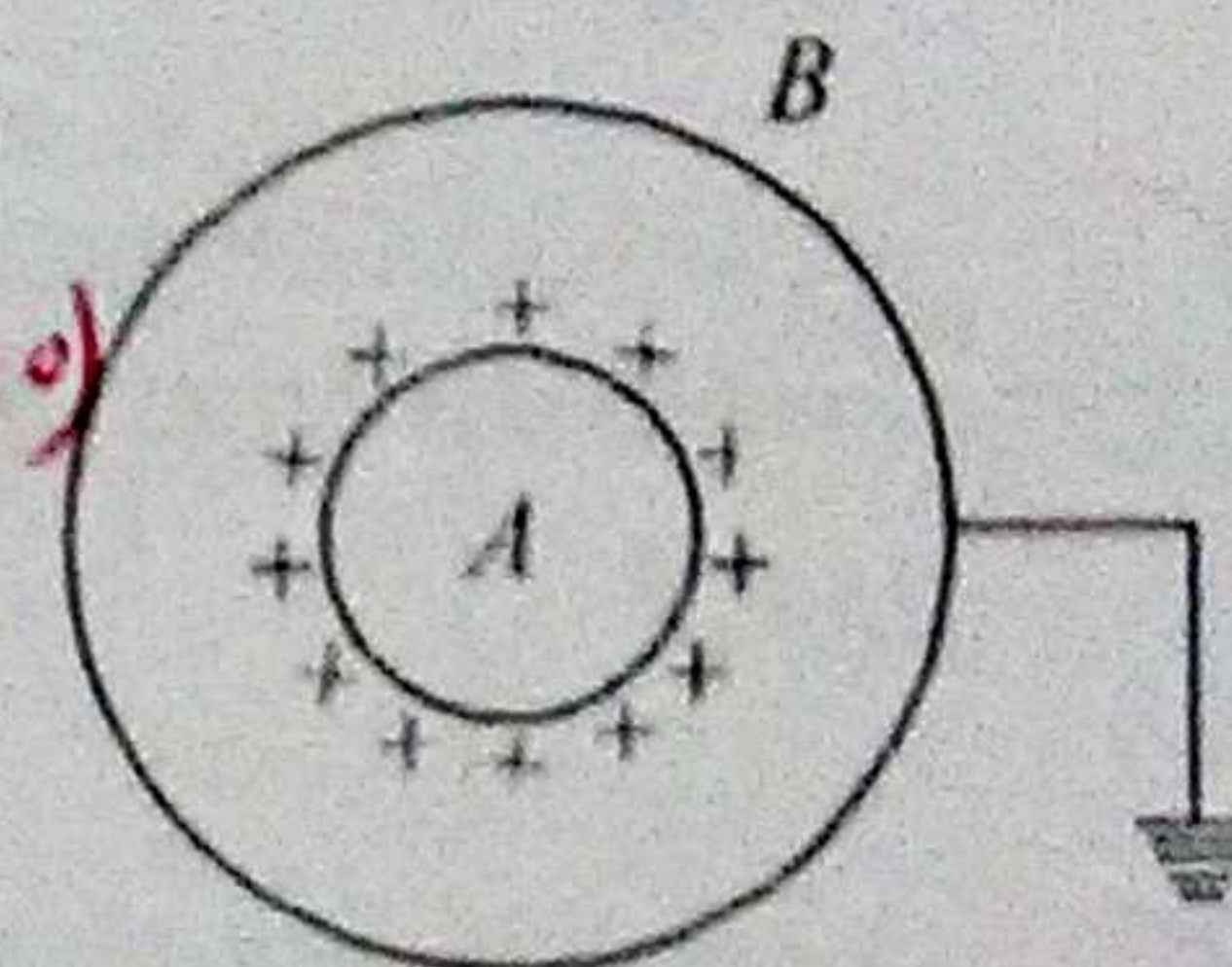


- (A) $\frac{2Q}{\epsilon_0 A}$ (B) $\frac{Q}{2A\epsilon_0}$ towards left (C) $\frac{Q}{2A\epsilon_0}$ towards right (D) $\frac{Q}{A\epsilon_0}$ towards right

3. A charge Q is uniformly distributed over a large plastic plate. The electric field at a point P close to the centre and just above the surface of the plate is 10 V/m . If the plastic plate is replaced by a copper plate of the same geometrical dimensions and carrying the same uniform charge Q , the electric field at the point P will become : *(charge will)*
- (A) Zero (B) 5 V/m (C) 10 V/m (D) 20 V/m

4. A and B are two conducting concentric spherical shells. A is given a charge Q while B is uncharged. If now B is earthed as shown in figure. Then :

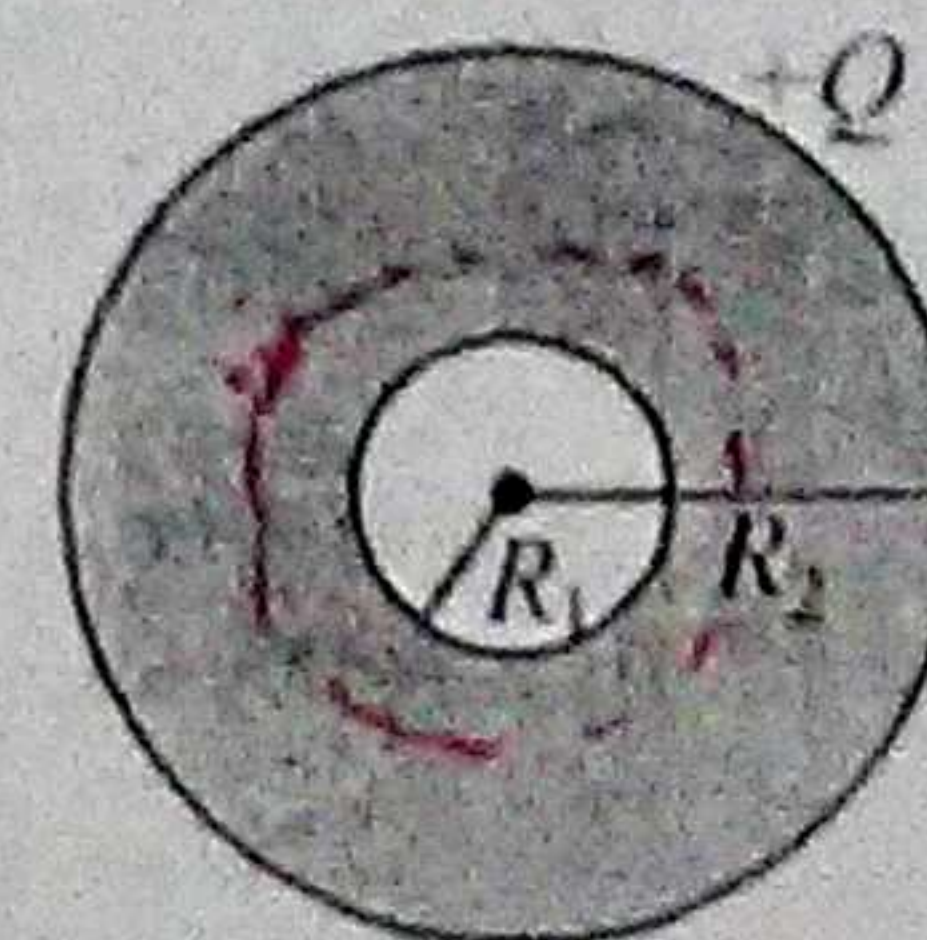
(write potential on $B=0$)



- (A) The charge appearing on inner surface of B is $-Q$
 (B) The field inside and outside A is zero
 (C) The field between A and B is not zero
 (D) The charge appearing on outer surface of B is zero

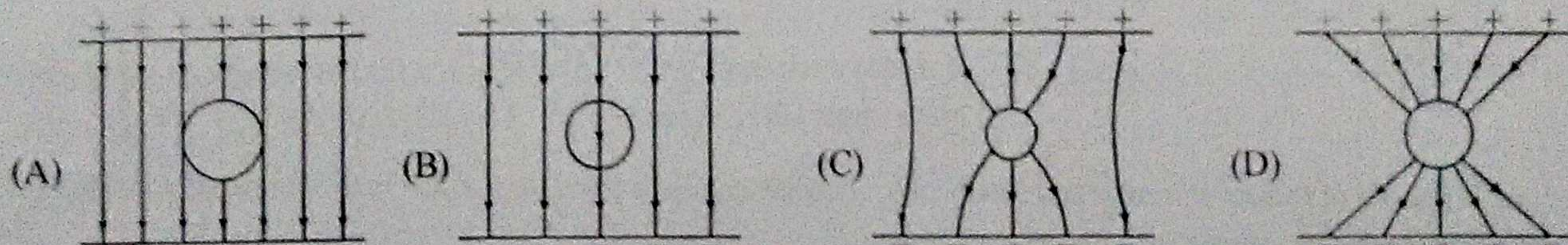
5. Figure shows a thick metallic sphere. If it is given a charge $+Q$, then electric field will be present in the region :

(charge will come to surface)

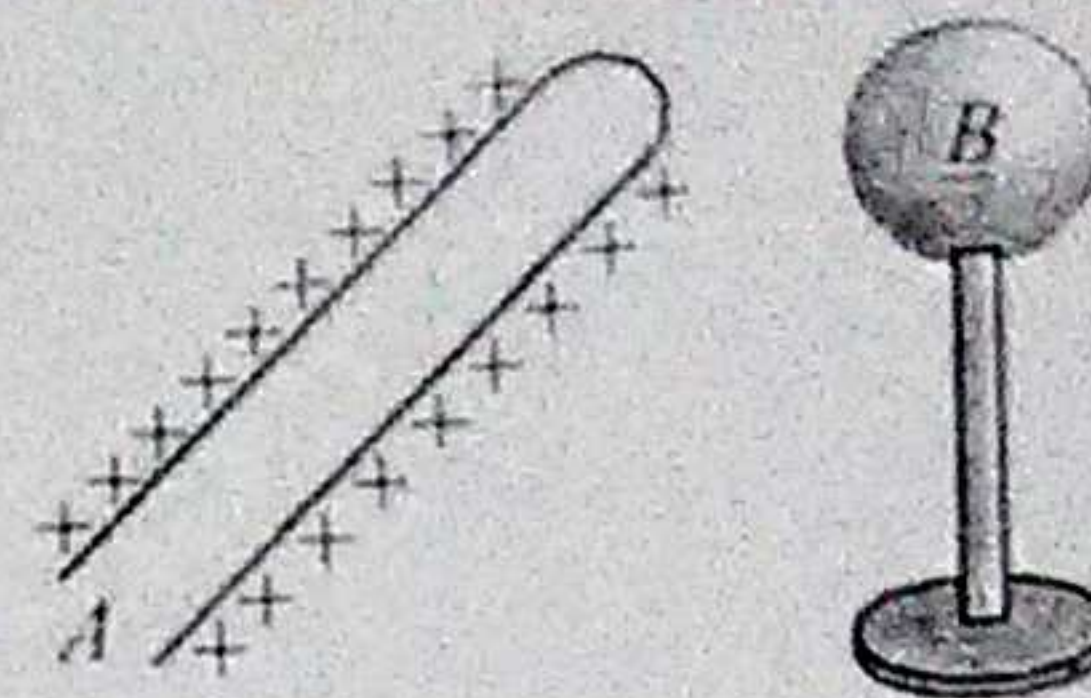


- (A) $r < R_1$ only
 (B) $r > R_1$ and $R_1 < r < R_2$
 (C) $r \geq R_2$ only
 (D) $r \leq R_2$ only

6. An uncharged sphere of metal is placed in a uniform electric field produced by two large conducting parallel plates having equal and opposite charges, then lines of force look like :

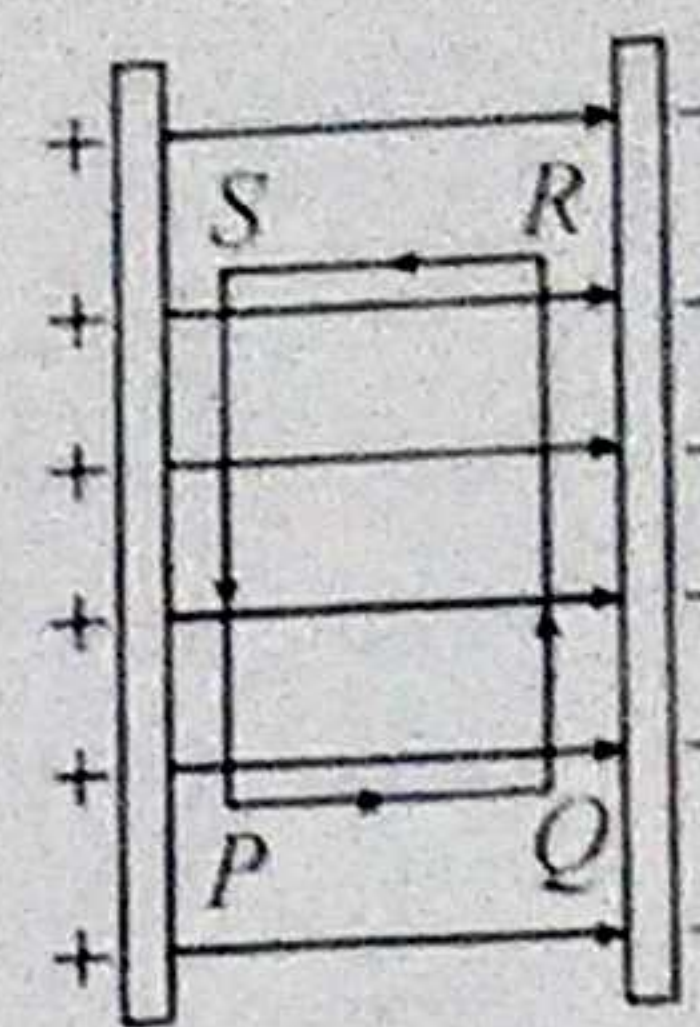


7. A positively charged body A has been brought near a neutral brass sphere B mounted on a glass stands as shown in the Figure. The potential of B will be :
- (A) Zero
(B) Negative
(C) Positive
(D) Infinite

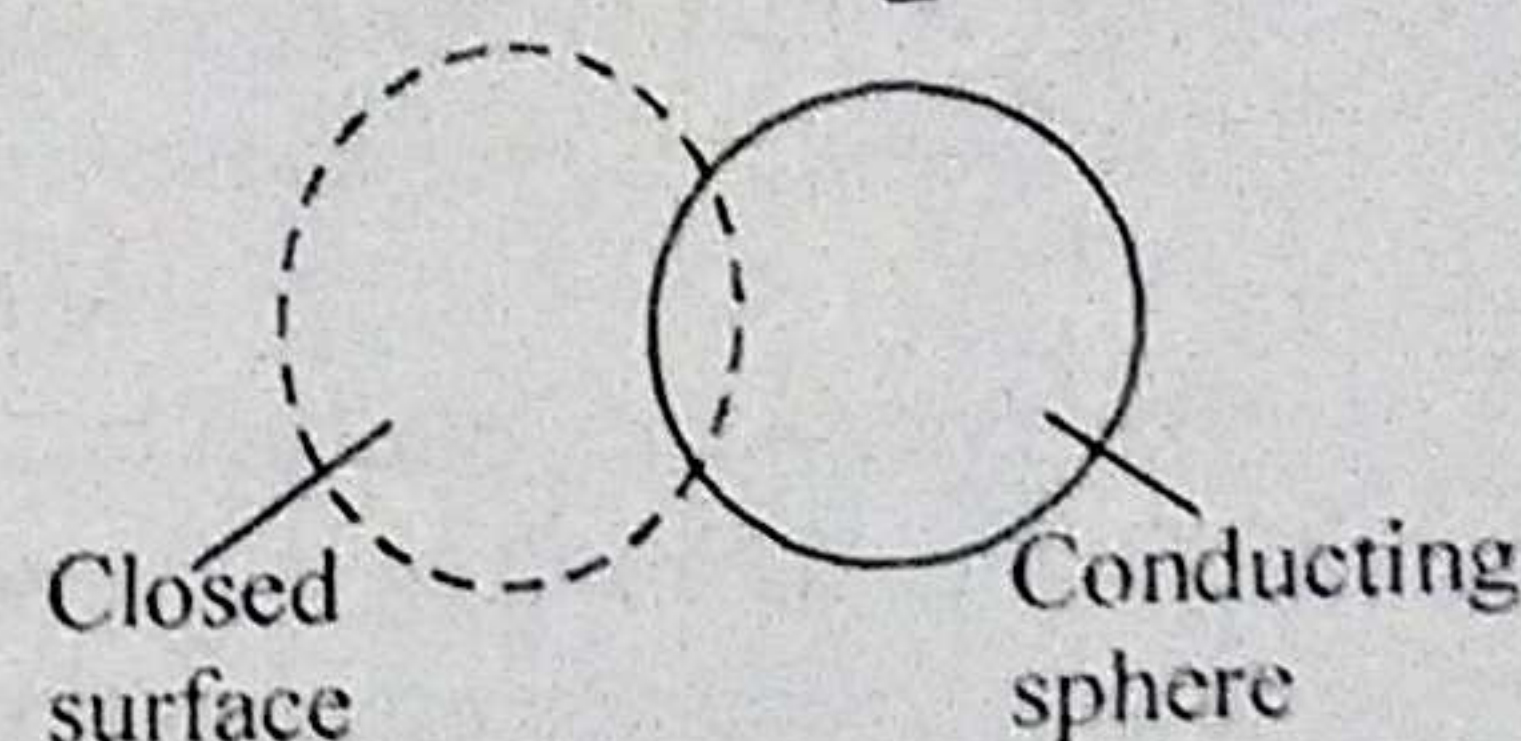


8. The amount of work done in Joules in carrying a charge $+q$ along the closed path $PQRSP$ between the oppositely charged metal plates is (Where E is electric field between the plates) :
- (A) Zero
(B) q
(C) $qE(PQ + QR + SR + SP)$
(D) q/ϵ_0

(Conservative Field)

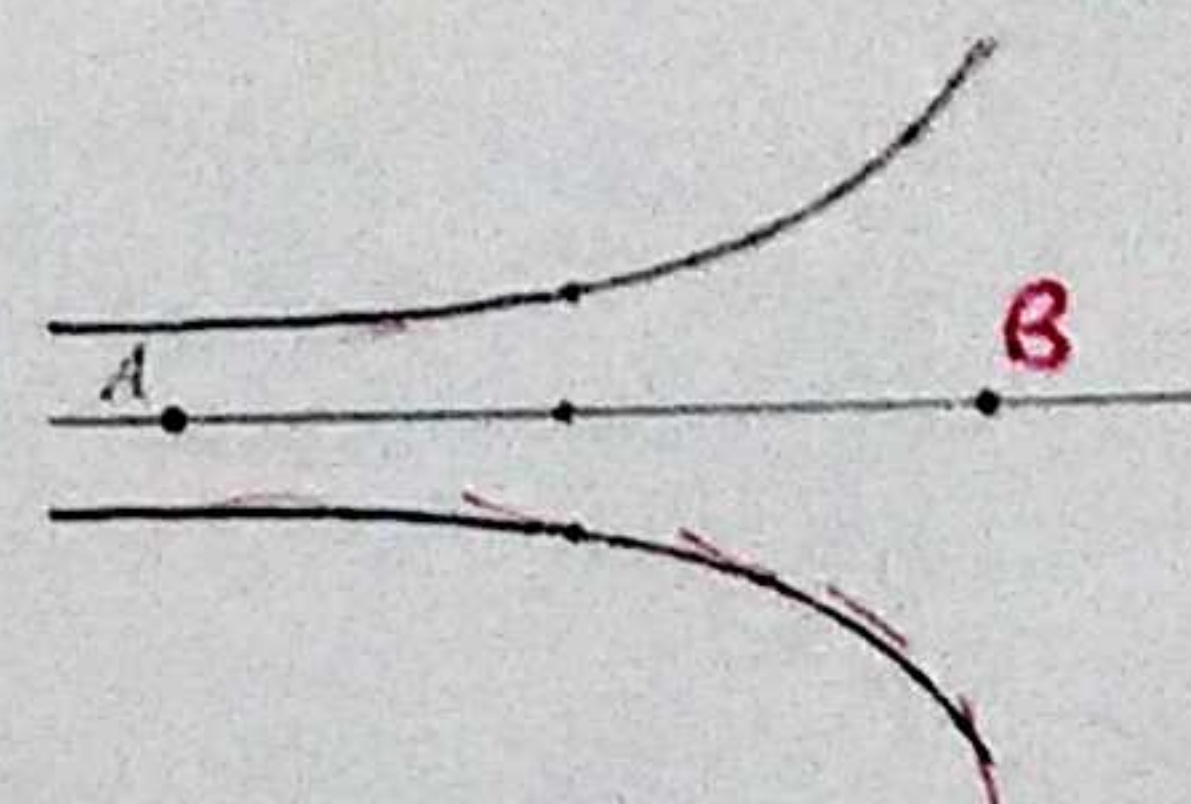


9. Figure shows a closed surface which intersects a conducting sphere. If a positive charge is placed at the point P , the flux of the electric field through the closed surface :
- (A) Will remain zero
(B) Will become positive
(C) Will become negative
(D) Will become undefined



10. In a solid uniformly charged sphere of total charge Q and radius R if energy stored outside the sphere is U_0 joules then find out self energy of sphere in term of U_0 ?

11. Figure shows some of the electric field lines corresponding to an electric field. The figure suggests that (E = electric field, V = potential) :
- (A) $E_A > E_B$
(B) $E_A < E_B$
(C) $V_A > V_B$
(D) $V_A < V_B$



12. A particle A of mass m and charge Q moves directly towards a fixed particle B , which has charge Q . The speed of A is v when it is far away from B . The minimum separation between the particles is proportional to :
- (A) Q^2
(B) $\frac{1}{v^2}$
(C) $\frac{1}{v}$
(D) $\frac{1}{m}$

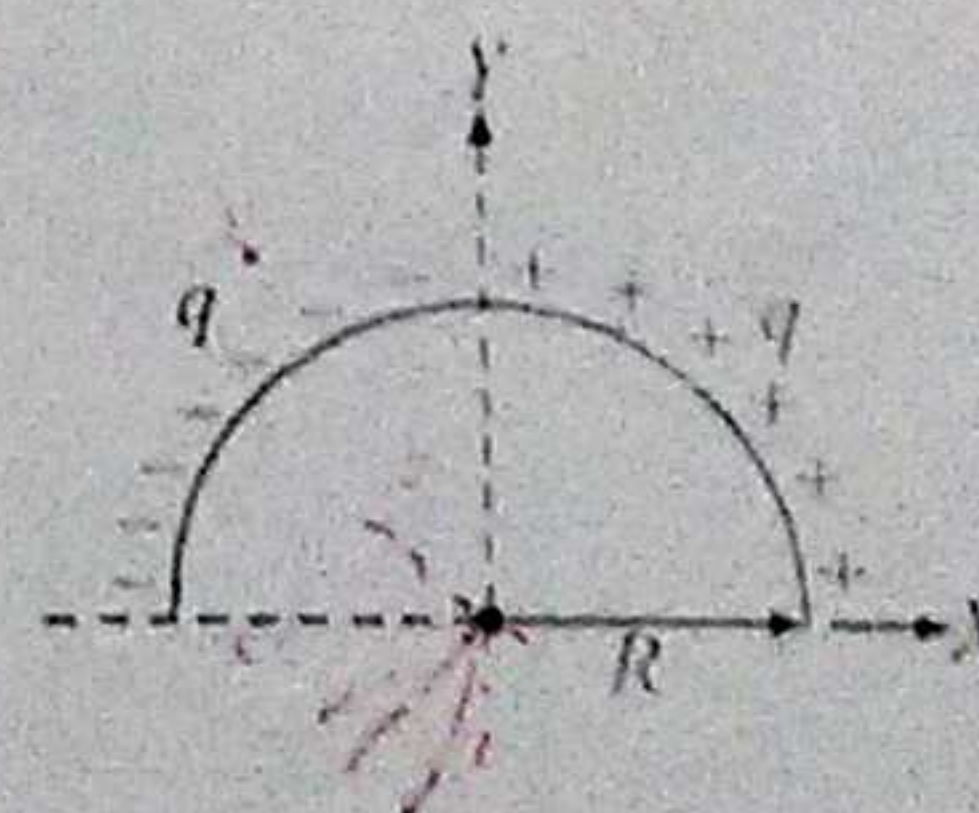
13. A point charge q is placed at origin. Let $\vec{E}_A$, $\vec{E}_B$ and $\vec{E}_C$ be the electric field at three points $A(1, 2, 3)$, $B(1, 1-1)$ and $C(2, 2, 2)$ due to charge q . Then :
- (A) $\vec{E}_A \perp \vec{E}_B$
(B) $\vec{E}_A \parallel \vec{E}_C$
(C) $|\vec{E}_B| = 4|\vec{E}_C|$
(D) $|\vec{E}_B| = 8|\vec{E}_C|$

14. A spherical conductor A lies inside a hollow spherical conductor B . Charges Q_1 and Q_2 are given to A and B respectively :
- (A) Charge Q_1 will appear on the outer surface of A
(B) Charge $-Q_1$ will appear on the inner surface of B
(C) Charge Q_2 will appear on the outer surface of B
(D) Charge $Q_1 + Q_2$ will appear on the outer surface of B

15. Electric field at centre of semicircular ring shown in figure is :

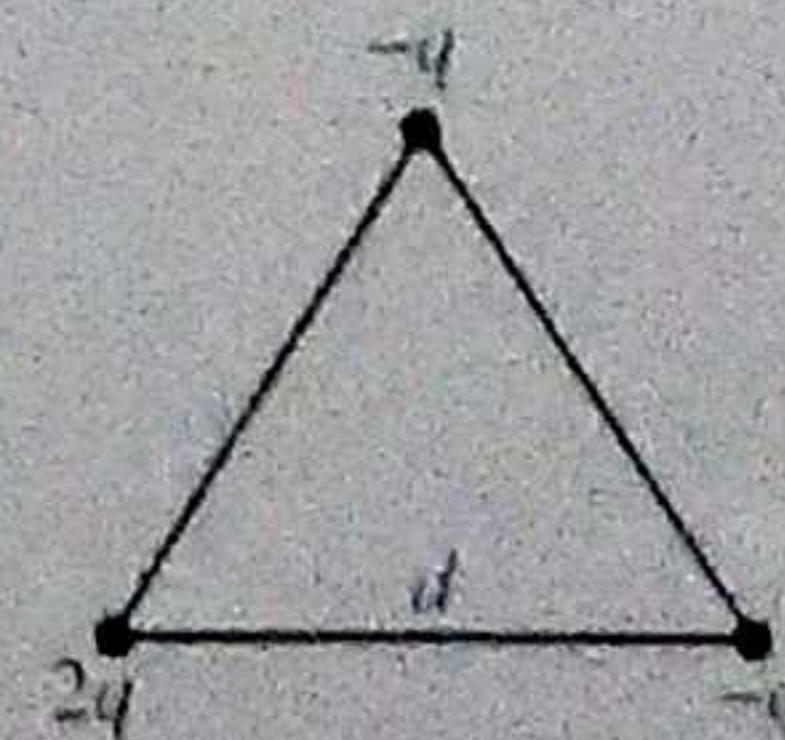
(A) $\frac{4Kq}{\pi R^2} \hat{j}$
(C) $-\frac{2\sqrt{2}Kq}{\pi R^2} \hat{i}$

(B) $-\frac{4Kq}{\pi R^2} \hat{i}$
(D) $-\frac{2\sqrt{2}Kq}{\pi R^2} \hat{j}$

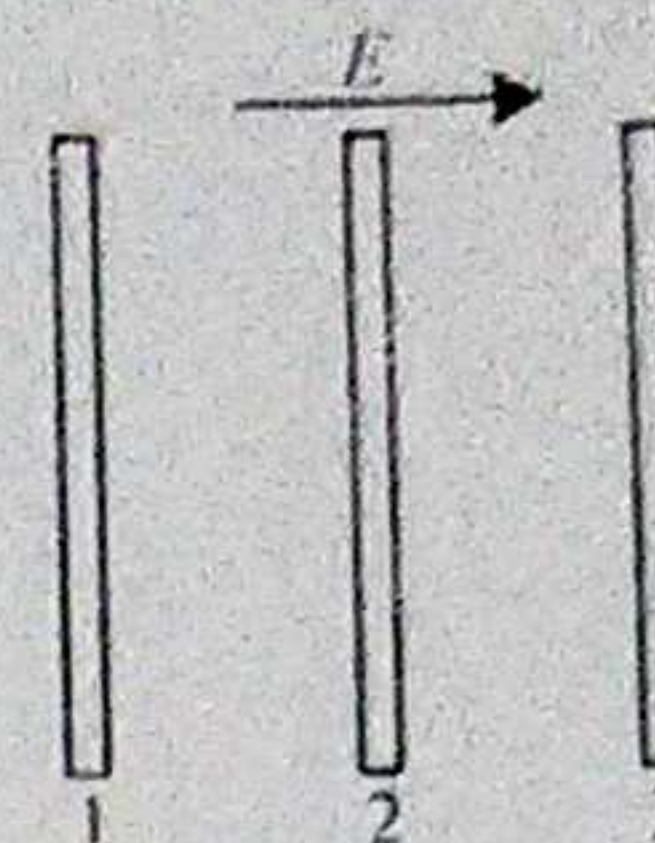


16. A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre ?
- (A) R
(B) $R/2$
(C) $R/3$
(D) $2R$

17. Three charges are arranged on the vertices of an equilateral triangle as shown in figure. Find the dipole moment of the combinations.



18. Three uncharged conducting large plates are placed parallel to each other in a uniform electric field. Find the induced charge densities on each surface of each plate.

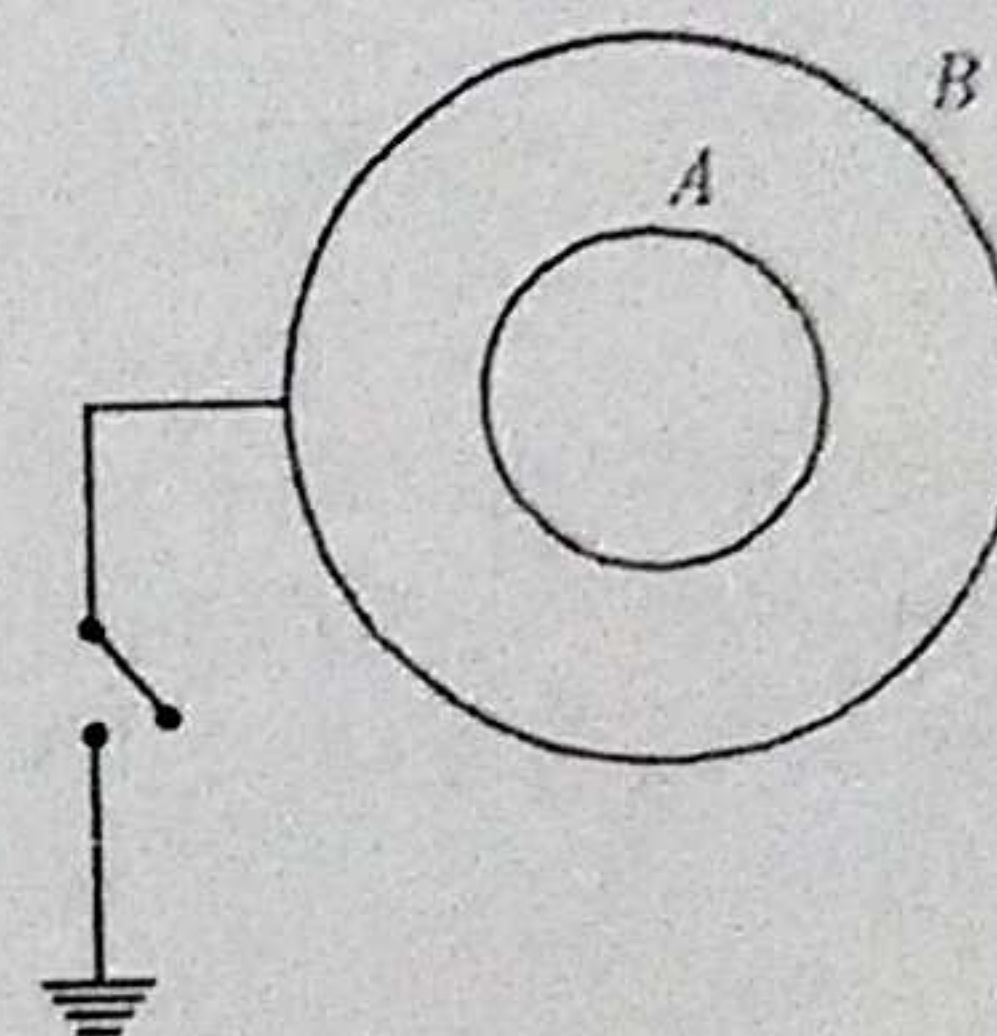


19. To move a point charge from one point to another point in an electric field :
 (A) Work done depends on path
 (B) Work done is independent of path
 (C) Work done may be dependent on path
 (D) No work is done in some cases and independent of path in other cases

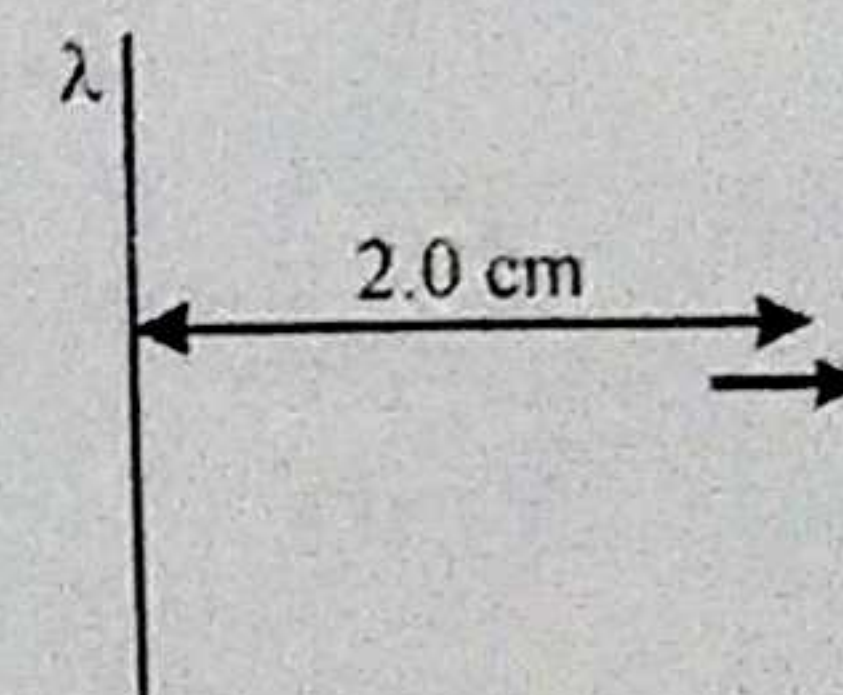
20. Ten charges (5 of them are $+q$ each & other five are $-q$ each) are placed randomly on the circumference of a circle. The radius of the circle is R . The electric potential at the centre of this sphere due to these charges will be :
 (A) 0 (B) $10 Kq/R$ (C) $5 Kq/R$
 (D) Cannot be calculated unless the positions of the charges on the circle are specified.

21. A copper ball 2 cm in diameter is immersed in a liquid (relative permittivity 5) of density 800 kg/m^3 . The electric field of intensity 400 V/m directed vertically upward is applied and a charge Q is given to the ball so that the ball remains suspended in the field. The value of the charge Q is ($\rho_{\text{cu}} = 8600 \text{ kg/m}^3$)
 (A) $8 \times 10^{-6} \text{ C}$ (B) $4 \times 10^{-6} \text{ C}$ (C) $4 \times 10^{-3} \text{ C}$ (D) $4 \times 10^{-4} \text{ C}$

22. Two concentric shells have radii R and $2R$ charges q_A and q_B and potentials 2 V and $(3/2) \text{ V}$ respectively. Now shell B is earthed and let charges on them become q'_A and q'_B . Then :
 (A) $q'_A/q'_B = 1/2$
 (B) $q'_A/q'_B = 1$
 (C) Potential of A after earthing becomes $(3/2) \text{ V}$
 (D) Potential difference between A and B after earthing becomes $V/2$



23. An electric dipole consists of charges $\pm 2.0 \times 10^{-8} \text{ C}$ separated by a distance of $2.0 \times 10^{-3} \text{ m}$. It is placed near a long line charge of linear charge density $4.0 \times 10^{-4} \text{ C/m}$ as shown in figure, such that the negative charge is at a distance of 2.0 cm from the line charge. Find the force acting on the dipole.



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ANSWER KEY

1. (D) 2. (C) 3. (C) 4. (A)(C)(D) 5. (C) 6. (C) 7. (C) 8. (A) 9. (B)
 10. $[\frac{6U_0}{5} \text{ Joules}]$ 11. (A)(D) 12. (A)(B)(D) 13. (A)(C) 14. (A)(B)(D) 15. (B) 16. (C)
 17. $[q d \sqrt{3}]$, along the bisector of the angle at $2q$, away from the triangle] 18. $[-\epsilon_0 E, \epsilon_0 E \dots \dots \text{ and so on}]$
 19. (B) 20. (A) 21. (C) 22. (A)(D) 23. $[0.6 \text{ N towards the line charge}]$